

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (Civil) (Credit System) Degree Examinations

Supplementary Examination - May - June 2023

20CV504 - TRANSPORTATION ENGINEERING - I

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Assume any missing data suitably as per IRC.

3) Use of provided copy of IRC charts is allowed.

Unit – I

Marks BT* CO* PO*

1. a) Compare the characteristic features of different modes of transportation.

10 L*2 1 1

- b) (i) Explain the objectives and requirements of providing transition curves in the horizontal alignment.

(ii) Design the rate of super elevation for a horizontal National Highway curve of radius 245 m and the design speed of 90 kmph.

10 L3 2 2

2. a) Work out the utility per unit length of each system and indicate which system is considered to be optimum with maximum utility per unit length. Assume utility units of 0.25, 0.50, 1.0 and 2.5 respectively for various population ranges; and utility units of 1.0 for every hundred tons of agricultural and Industrial productivity

Proposal		P	Q	R	S	T
Total Road Length (km)		300	400	500	550	590
Number of villages served with population range	<1000	160	200	240	248	285
	1000-2000	80	90	110	112	182
	2000-5000	30	60	70	73	81
	>5000	6	8	10	12	19
Total agriculture and Industrial productivity served in thousand tons		200	270	315	335	298

10 L2 1 2

- b) Define sight distance. List and explain the different types of sight distances considered in geometric design of highways.

10 L3 2 1

3. a) (i) Sketch (x) star and circular pattern (y) Block pattern of road networking.

(ii) Explain the methods of classification of roads.

10 L2 1 1

- b) An ascending gradient of 1 in 120 meets a descending gradient of 1 in 150. Identify the vertical curve and design the length of the same as per IRC guidelines for a design speed of 100 kmph for both stopping and overtaking sight distance requirements. Assume the overtaking sight distance as 500 m.

10 L3 2 3

Unit – II

4. a) Define modulus of subgrade reaction and deduce the modulus of subgrade reaction of soil from the following test results adopting graphical method: A plate load test was conducted on a soaked subgrade during monsoon season using a plate of diameter 30 cm. The load values corresponding to the mean settlement dial

P.T.O.

10 L3 3 1

readings are given below.

Settlement (mm)	Load (kg)
0	0
0.24	460
0.52	900
0.76	1180
1.02	1360
1.23	1480
1.53	1590
1.76	1640

- b) (i) Write a note on the significance of Central Road Fund in Indian Highway Financing.
(ii) Enumerate any six factors to be considered for evaluating vehicle operation cost. 10 L2 4 1
5. a) Clearly define the following terms pertaining to traffic engineering "Traffic speed", "Traffic volume", "Traffic capacity", "Traffic density" and "PCU". Indicate the units associated with each. 10 L2 3 1
b) A two-lane two way road is at present carrying a traffic of 1200 CVPD. It is to be strengthened for the growing traffic needs. The VDF was found to be 4.0. The rate of growth of traffic is 10% per annum. The total period of construction is 4 years. The pavement is designed for a life of 20 years. CBR of soil subgrade can be taken as 6.0%. Calculate the cumulative number of standard axles and design the flexible pavement section using IRC method. Show the design composition. 10 L3 4 .3
6. a) List and explain any five desirable properties and requirements of the road aggregates. Indicate the tests to be performed to measure each property. 10 L2 3 1
b) The details of three alternate proposals for strengthening of an existing highway pavement are given below. The present traffic consists of 5500 motor vehicles per day, with annual growth rate of 6.0 %. Determine which alternative is most economical, if the rate of interest payable in all the cases is 10% per annum.

Sl. no.	Overlay type	Design life, years	Construction cost, Rs. (lakhs) per km	Average maintenance cost, per km, Rs. (during design life)	Vehicle operation cost, Rs. per veh-km (during design life)
(i)	WBM + PMC	5	55.00	3,50,000	4.0
(ii)	BM + SDBC	8	69.51	7,50,000	3.5
(iii)	DBM + BC	12	95.00	1,00,000	3.0

10 L4 4 2

Unit – III

7. a) Illustrate the classification (types) of bridges. Explain any two types of bridges with neat sketches. 10 L2 5 1
b) (i) List any four basic objectives of highway maintenance.
(ii) Enumerate any six requirements of highway drainage system. 10 L2 5 1
8. a) Describe the structural deterioration of flexible and rigid pavements. 10 L2 5 1
b) With neat sketches explain the following types of failures in pavements (i) Map cracking (ii) Reflection cracking. 10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
